

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

Anonymous authors
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ABSTRACT

Medical image analysis depends on transferable visual representations, yet expert annotation remains scarce, costly, and difficult to standardize across clinical imaging modalities. Self-supervised masked image modeling (MIM) offers a direct response to this bottleneck, but dominant formulations still depend on random, high-ratio masking that expends reconstruction capacity on regions only weakly related to diagnostic anatomy or pathology. Existing region-aware alternatives improve selectivity only partially, because they often rely on reconstruction-error heuristics or supervised localization signals that constrain their adaptability across modalities and downstream objectives. We present Mask What Matters (MWM), a controllable text-guided masking framework for self-supervised medical image analysis. MWM uses vision-language models to extract prompt-conditioned localization cues and translates those cues into differentiated masking: diagnostically relevant regions receive concentrated reconstruction pressure, whereas semantically redundant background is masked more lightly. This design converts open-vocabulary textual intent into an explicit pretraining signal, thereby aligning masked reconstruction with the semantic structure of the evaluated medical imaging tasks. Across brain MRI, chest CT, and lung X-ray experiments, MWM consistently improves over representative MIM baselines, including SparK, with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection and instance segmentation. The same advantage is achieved at substantially lower overall masking ratios, exemplified by 40% for MWM versus 70% for SparK in the masking-ratio comparison. These findings establish controllable, text-driven masking as an effective mechanism for semantically aligned self-supervised representation learning in medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection are the principal predictive tasks through which computer vision systems support medical image analysis. Deep learning has markedly advanced these tasks, but its strongest empirical performance commonly depends on large annotated datasets. In medical imaging, where expert labels are expensive, time-consuming, and often difficult to standardize across modalities, this dependence remains a central barrier to scalable representation learning.

Self-supervised learning (SSL) directly mitigates this bottleneck by pretraining models on unlabeled data and transferring the resulting representations to annotated downstream tasks [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among SSL paradigms, masked image modeling (MIM) has become a particularly influential generative strategy: by hiding image regions and reconstructing them from visible context, MIM encourages encoders to capture structural regularities and semantic dependencies. This paradigm has achieved strong transfer performance in natural-image settings and has been adapted successfully to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

Dominant MIM formulations, including masked autoencoders (MAE) [He et al., 2022] and SparK [Tian et al., 2023], nevertheless select masked image content largely at random. Because random masking is agnostic to the regions that determine downstream objectives, it creates two coupled limitations. First, the reconstruction task can devote substantial capacity to visual content that is not semantically decisive for transfer. Second, to increase the chance that informative regions are hidden, MIM methods often adopt very high masking ratios, such as 75% in MAE, thereby increasing reconstruction difficulty and resource demand. This regime contrasts with masked language modeling, where the compact and semantically dense structure of text enables effective learning under lower masking rates [Devlin et al., 2019]. The reliance on high-ratio image masking therefore reflects not only visual redundancy but also the absence of a mechanism that selectively emphasizes the regions that matter.

This mismatch is especially consequential in medical imaging. Diagnostic evidence is frequently spatially sparse, appearing in localized lesions, organs, or anatomical structures, whereas large portions of an image may be clinically redundant for a given task. At the same time, medical imaging spans MRI, CT, and X-ray modalities whose task-relevant regions differ markedly in shape, scale, and spatial distribution. An effective masking strategy must therefore be selective enough to align pretraining with the downstream semantic target while remaining flexible across heterogeneous imaging conditions.

We address these challenges with **Mask-What-Matters (MWM)**, a controllable, text-guided masking framework for medical image MIM. Given a user-specified open-vocabulary phrase or description, a pretrained vision-language model provides zero-shot localization cues that are converted into robust regions of interest (ROIs). MWM then performs prompt-conditioned, region-specific masking: semantically important areas, such as lesions or organs, receive higher masking pressure, whereas background regions are masked at lower ratios. In this way, MWM injects downstream semantic intent into self-supervised pretraining without requiring per-image reports or labels.

Our contributions are threefold:

- We introduce controllable text-guided masking for medical image MIM. MWM integrates vision-language localization (BiomedCLIP) with segmentation refinement (SAM) to derive task-relevant regions from open-vocabulary prompts and to apply differentiated ROI versus background masking, directly addressing the semantic misalignment of random masking.
- We provide an annotation-free, backbone-agnostic pretraining design. MWM requires no per-image reports or labels and can be integrated into both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- We demonstrate consistent empirical gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM outperforms representative MIM baselines, including SparK, improving classification, detection, and segmentation performance within the evaluated settings.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first established in natural language processing by bidirectional encoder representations from transformers (BERT) [Devlin et al., 2019], which learns contextual representations by masking tokens and predicting them during pretraining. [He et al., 2022] transferred this principle to computer vision through MIM, where image regions are hidden and reconstructed from the remaining visual context. The resulting paradigm has shown strong generalization in both natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative methods include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]: MAE uses high-ratio random masking to train ViTs toward higher-level semantic representations; MedMAE adapts the paradigm to medical images; and SparK extends MIM to convolutional backbones by exploiting multi-scale feature hierarchies.

Despite this progress, conventional MIM typically remains random or structure-agnostic at the masking stage and therefore does not explicitly encode the semantic requirements of downstream tasks. Non-selective masking can cause pretraining to overemphasize irrelevant regions and weaken

transfer effectiveness. Recent region-aware methods seek to correct this alignment problem, with two principal directions emerging.

The first direction uses loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al. (2023a)] and AnatoMask [Li et al. (2024)] analyze regional reconstruction errors and assign higher masking probabilities to high-loss areas, aiming to prioritize semantically valuable structures. This criterion, however, can conflate reconstruction difficulty with task relevance. In brain magnetic resonance imaging (MRI), for example, complex structures such as the spinal cord or eyeballs may attract masking attention even when they are unrelated to tumor-oriented downstream tasks.

The second direction incorporates external perceptual modules. FocusMAE [Basu et al. (2024)], for instance, uses pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although such modules can be effective in specific tasks, they require supervised RPN training on domain-specific annotated data, limiting their flexibility when tasks or modalities change.

These limitations motivate a masking mechanism that is semantically aligned, flexible across tasks, and independent of per-image supervision. Natural language provides precisely this kind of compact and interpretable guidance, making it well suited for task-aware MIM when it can be linked reliably to image regions.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Recent progress in vision-language alignment has made zero-shot and open-vocabulary localization increasingly practical [Radford et al. (2021)]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared embedding space for images and text through large-scale image-text pretraining and has become a foundation for zero-shot and open-set visual recognition. In medical imaging, BiomedCLIP [Zhang et al. (2023)] strengthens cross-modal understanding for biomedical content and supports robust performance across downstream applications. Most uses of vision-language pre-trained models nevertheless remain concentrated in supervised downstream tasks, including object detection and semantic segmentation [Zhong et al. (2022), Aleem et al. (2024), Koleilat et al. (2024)]. Their open-vocabulary and zero-shot capabilities have not been fully exploited as mechanisms for controlling the MIM pretraining process itself.

Taken together, prior region-aware MIM methods improve semantic focus but remain constrained by additional supervision, imperfect alignment between masking criteria and task semantics, or limited adaptability. Vision-language models, by contrast, offer a direct route from textual task intent to visual localization. MWM connects these two lines of work by using natural language prompts to determine where MIM should allocate reconstruction pressure during pretraining.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** consists of two coupled stages: text-guided region localization and region-aware masked image modeling. The localization stage uses a frozen vision-language model to map an open-vocabulary prompt to task-relevant image regions. The pretraining stage assigns differentiated masking ratios to those regions and to the remaining background, making the reconstruction objective explicitly responsive to the semantic target specified by the prompt.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM obtains task-relevant regions through a structured localization pipeline. It first constructs textual prompts and uses BiomedCLIP to embed the prompt and image in a shared cross-modal space. A Multi-Modal Information Bottleneck (M2IB) module then produces a prompt-conditioned saliency map, which is binarized with K-Means and refined by the Segment Anything Model (SAM). The refined mask is finally converted into a bounding box with a controllable expansion margin, yielding the spatial guide used by the masked modeling stage.

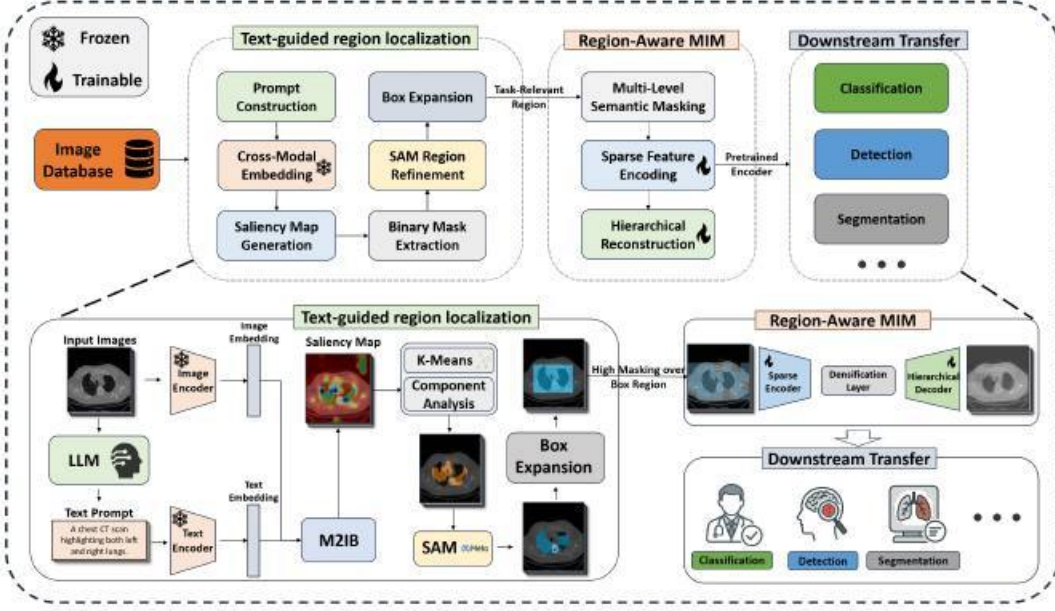


Figure 1: Overview of the MWM framework. The top panel summarizes the complete workflow: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel specifies how open-vocabulary prompts are converted into localization cues and differentiated masking decisions.

Prompt Generation. Textual prompts define the semantic target of localization and consequently determine which image regions are emphasized during pretraining. To obtain descriptive guidance, large language models (LLMs) are queried with structured instructions such as “Describe the typical visual characteristics of a [category] in a [modality] image”. We also evaluate simple category phrases as alternative prompts, enabling a controlled comparison between concise and descriptive language under the same localization and pretraining pipeline.

Cross-Modal Embedding. Given a prompt, BiomedCLIP [Zhang et al. (2023)] encodes the medical image and text into a shared semantic space. For an image I and prompt T , the corresponding embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These embeddings define the cross-modal representation on which prompt-conditioned saliency estimation is based.

Saliency Map Generation. A cross-modal saliency module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)] localizes task-relevant visual evidence by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB optimizes the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the BiomedCLIP image and text embeddings, respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering task-irrelevant visual information.

Binary Mask Extraction. The saliency map is binarized with unsupervised K-Means clustering [Lloyd (1982)] to produce an initial prompt-guided foreground mask. The map is partitioned into two pixel groups, and the group with higher saliency values is treated as foreground. Connected component analysis is then applied to the binary map to retain the largest foreground regions and suppress small false activations, especially along noisy boundaries.

Region Refinement. SAM is used as a spatial refinement module to improve the geometric fidelity of the localized region Kirillov et al. (2023). For each selected connected region $c_i \in \mathcal{C}^*$, the minimal enclosing bounding box is computed and supplied as the visual prompt for SAM.

Given the original image I and the corresponding bounding boxes, SAM predicts the refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} preserves a more faithful spatial structure than the coarse saliency-derived region.

Box-Based Region Expansion. Because the SAM output may still contain boundary irregularities, direct use of the mask can introduce unstable spatial guidance during representation learning. MWM therefore converts the refined SAM mask into a bounding box with a controllable expansion margin. This expanded box defines the final region of interest (ROI) for the masked modeling procedure.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the ROI produced by the localization module in Section 3.2, MWM performs masked image modeling through three operations: multi-level semantic masking, sparse feature encoding, and hierarchical reconstruction. The masking operation assigns ratios according to region importance, the encoder processes only visible patches, and the decoder reconstructs the image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM uses the localized semantic regions to define a multi-level masking pattern. Patches inside prompt-highlighted regions, such as tumor- or organ-related areas, are masked at a higher ratio so that reconstruction pressure concentrates on task-relevant structures. Background patches are masked lightly or left visible, reducing redundant reconstruction of low-value image content. Figure 2 visualizes the resulting masking behavior.

Sparse Feature Encoding. Visible patches are assembled into a sparse image and encoded following the SparK design Tian et al. (2023):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i denotes the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ denotes the resulting multi-scale features. Although our implementation instantiates MWM with convolutional backbones, the same interface applies to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image through a top-down feature hierarchy. At each scale l , masked locations are densified with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , which is then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The reconstruction objective is the mean-squared error over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

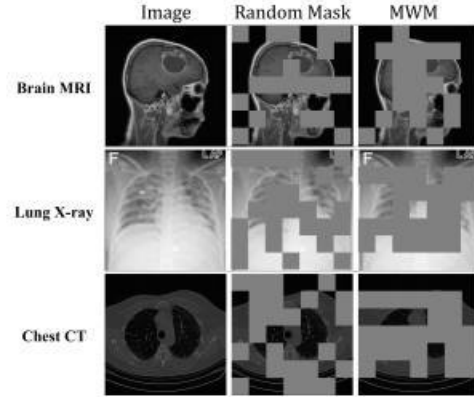


Figure 2: Comparison of masking strategies. Gray blocks denote masked regions.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining uses three datasets that span complementary medical imaging modalities. (1) *Brain MRI* includes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering various tumor types and anatomical structures. (2) *Lung X-ray* consists of around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation uses three datasets covering classification and detection-oriented transfer settings. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images spanning 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermary et al. (2018)] consists of 5,863 pediatric chest X-rays categorized as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are evaluated with backbones of comparable model scale: convolution-based methods adopt ResNet-50 [He et al. (2016)], whereas transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224 , and models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning retains the same encoder and input resolution, ensuring that performance differences reflect the pretraining strategy rather than changes in the transfer architecture. All experiments are performed on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables (e.g., prompt type or SAM usage) are fixed to their empirically optimal settings during evaluation to maintain controlled comparisons.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first evaluate whether the text-guided localization module produces spatial regions that correspond to expert annotations across brain MRI, lung X-ray, and chest CT. The analysis isolates two controllable factors—prompt type and SAM refinement—and compares predicted bounding boxes with expert-annotated segmentation masks. We report occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

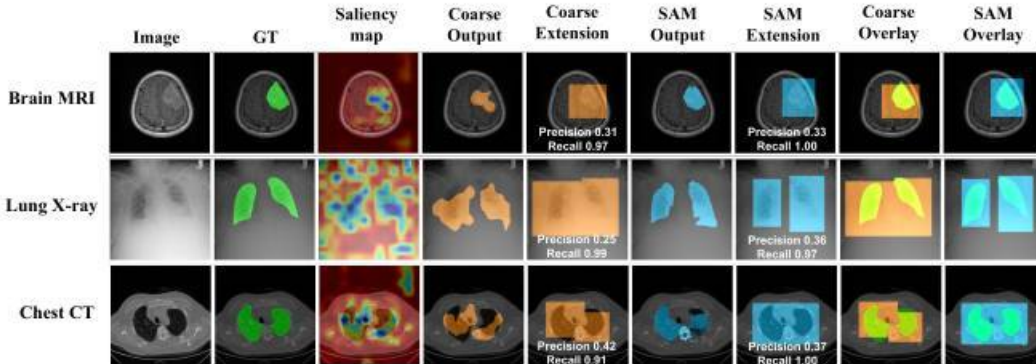


Figure 3: Representative text-guided localization results across medical imaging modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Text-guided localization performance measured by Occlusion Precision/Recall across prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* indicates further refinement using SAM.

As shown in Table 1, text-guided localization with effective prompts achieves consistently high occlusion recall across all datasets (around 0.82) while maintaining reasonable precision. This pattern confirms that the framework can identify task-relevant regions from textual descriptions alone, despite modality-specific variation in anatomy and image appearance (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. To assess the role of language specificity, we compare two prompt types under the same localization protocol. As shown in Table 1, descriptive sentence prompts yield stronger results for Brain MRI and Chest CT, indicating that richer semantic context improves localization when target structures are visually heterogeneous. In lung radiographs, concise phrases achieve higher recall (0.82 vs. 0.66), likely because the lungs occupy broad anatomical coverage for which general terms provide sufficient guidance.

Effect of SAM Refinement on Region Localization. We also quantify the contribution of spatial refinement by applying SAM to the coarse masks. As shown in Table 1, SAM refinement improves precision while preserving high recall. In Brain MRI, for example, precision increases from 0.22 to 0.30 while recall remains above 0.94. This result supports the use of SAM as a refinement stage for converting coarse prompt-conditioned localization into more precise spatial guidance.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the transfer quality of MWM by comparing it with representative self-supervised learning (SSL) methods from both reconstruction-based and contrastive learning paradigms. All methods are pretrained and evaluated with consistent datasets and training settings, so the comparisons directly assess the effect of the masking and pretraining strategy.

Image Classification. Classification performance is examined on three downstream medical imaging datasets. Following common practice, results are reported under two transfer protocols: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained on top.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets. MIM denotes Masked Image Modeling, and CL denotes Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with the encoder frozen.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM achieves the highest accuracy on all three datasets, demonstrating robust transfer across the evaluated imaging modalities. The largest gain appears on Chest CT, where MWM reaches 97.5% accuracy, exceeding SparK by +3.1 and AnatoMask by +1.7. This improvement highlights the value of text-guided masking, which concentrates reconstruction pressure on semantically critical yet spatially sparse regions—such as lung lobes in CT scans—while reducing redundant background reconstruction.
- **Linear probing:** Linear probing further evaluates the intrinsic quality of the learned encoder representations. For a focused comparison, MWM is evaluated against two recent masked image modeling methods, SparK and AnatoMask. As shown in Table 3, MWM outperforms both across all datasets, indicating that text-driven pretraining improves feature quality even when the encoder is not fine-tuned.

Object Detection and Instance Segmentation. To assess transfer beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As shown in Table 4, MWM achieves the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. These results show that the representations learned by MWM transfer effectively to both object-level localization and pixel-wise segmentation.

In summary, incorporating text-driven masking during pretraining yields more informative and transferable representations. The empirical pattern across classification, detection, and segmentation demonstrates that semantic guidance improves visual representation learning within the evaluated medical imaging settings.

4.5 ABLATION STUDY

We conduct ablation experiments to quantify how prompt design, SAM-based region refinement, and masking ratio each contribute to downstream classification performance in MWM.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas the opposite trend is observed on lung X-ray. This pattern is consistent with the localization results in Section 4.3: prompt forms that improve region identification also produce stronger downstream classification. Both prompt types, despite their different granularity, consistently outperform random masking across all datasets. This result confirms that text-guided masking provides effective semantic alignment between pretraining and downstream tasks. The SAM ablation further

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

shows that incorporating spatial refinement during pretraining consistently improves downstream classification, supporting its role in strengthening text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM achieves 96.8% classification accuracy at a masking ratio of 40%, a substantially lower setting than the roughly 70% ratios commonly used in masked image modeling. Even at this reduced ratio, MWM exceeds SparK’s best performance at 70% masking by +2.4% absolute accuracy. To rule out the possibility that SparK’s inferior performance reflects a suboptimal masking configuration, SparK is additionally evaluated under 40%, 50%, 60%, and 70% masking ratios. The results confirm that 70% masking yields the highest accuracy for SparK, strengthening the comparison. These results challenge the assumption that high masking ratios are inherently optimal for MIM. Instead, they show that semantically guided, task-aware masking can achieve stronger downstream performance with lower masking ratios by preserving task-relevant content and reducing redundant reconstruction.

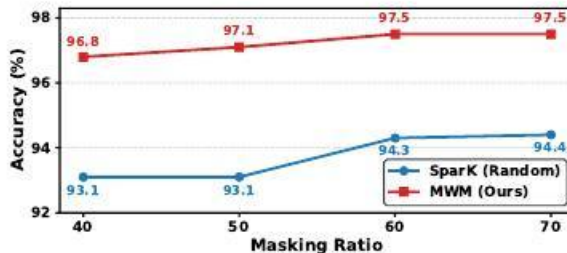


Figure 4: Classification performance on the Chest CT dataset under different masking strategies and ratios.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. By using natural language prompts to identify task-relevant areas and applying differentiated masking to those regions and the background, MWM enables semantically aligned representation learning without requiring per-image annotations.

The experiments substantiate three central findings: (1) natural language prompts reliably guide semantic region localization, validating text-driven masking as a feasible pretraining control mechanism; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across three imaging modalities and also improves detection and instance segmentation relative to SparK; (3) MWM maintains strong performance at lower masking ratios (e.g., 40%), underscoring the advantage of semantic guidance over uniformly high random masking in the evaluated settings.

Limitations. The framework also has limitations that define the scope of the present evidence. First, although text prompts provide flexible guidance, the robustness of MWM to variation in prompt style, vocabulary, or noise has not been systematically examined. Second, reliance on external textual descriptions introduces a dependency that may weaken generalization when reliable prompts are unavailable or inconsistent.

Looking forward, text-driven, region-aware masking offers a principled route toward more semantically grounded self-supervised learning. Extending this form of controllable masking beyond the present medical imaging setting may open new opportunities for vision–language-guided pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

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